Team Members:
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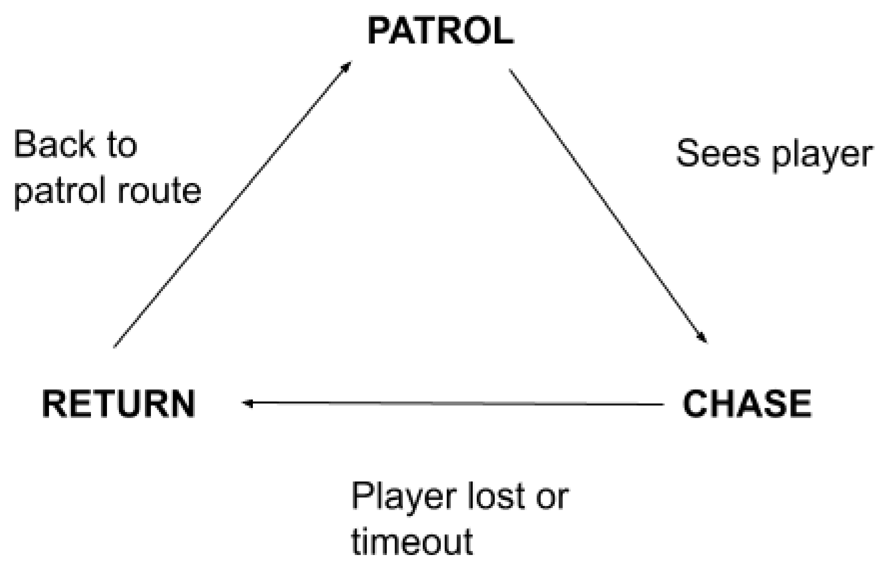
State Machine Diagram:



Ai Implementation:
The AI is implemented using a state machine with three core states: Patrol, Chase, and Return. The guard begins in Patrol(blue), moving between predefined waypoints. When the player enters detection range and is visible, the guard transitions to Chase (red with !). If the player is lost or moves out of range, the guard transitions to Return (yellow), where it moves back toward its patrol route. Once the guard reaches its patrol route, it resumes Patrol behavior. If the guard reaches the player within a defined catch range, the game ends.

Reflections:

Wendy: For the project I implemented the enemy AI (state machine). At the beginning of the project I thought that implementing AI would be hard. However, in Unity, implementing enemy AI was easier than I thought. It already provides built-in systems that make AI navigation easy. For example, with NavMesh, it handles a lot of the movement and navigation automatically instead of having to pathfinding completely from scratch.

Taylor: For this project I implemented the start/finish, the hiding spots, and the player. I also helped Wendy a little bit with the AI but it was mostly her. One thing I learned is that implementing AI with unity is actually not as hard as I expected it to be. I think this is due to the plethora of material out there that's related to utilizing AI in unity.